8.5: Partial Order Relations

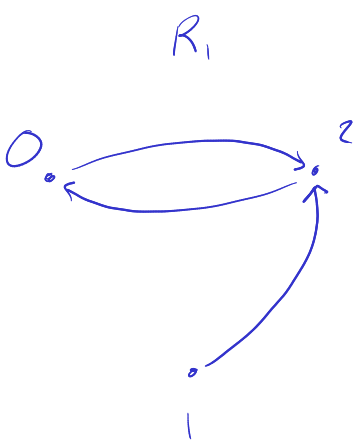
Definition.
Let R be a relation on a set A. R is **antisymmetric** if, and only if, for every a and b in A, if $a\,R\,b$ and $b\,R\,a$, then $a = b$.

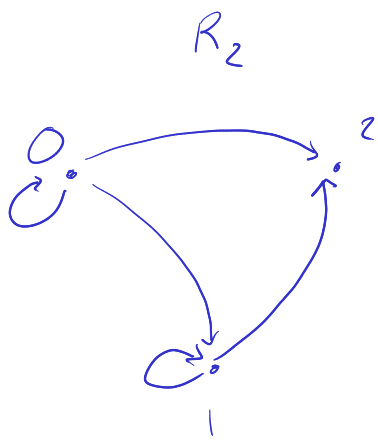
Alternatively, if $a \neq b$, then either $a \not R\, b$ or $b \not R\, a$.

Example. Let R_1 and R_2 be relations on $\{0, 1, 2\}$ defined as follows:

$$R_1 = \{(0,2), (1,2), (2,0)\}, \qquad R_2 = \{(0,0), (0,1), (0,2), (1,1), (1,2)\}$$

Draw the directed graphs for R_1 and R_2 and indicate which relations are antisymmetric.





Not antisymmetric

Antisymmetric

Definition.
Let R be a relation defined on a set A. R is a **partial order relation** if, and only if, R is reflexive, antisymmetric, and transitive.

Example. For non-empty sets U, and V, define the relation S such that

$$U\,S\,V \Leftrightarrow U \subseteq V$$

Prove that S is a partial order relation.

- Reflexive: $U R U$ because every set is a subset of itself.
- Antisymmetric: $U R V$ & $V R U \Leftrightarrow U \subseteq V$ & $V \subseteq U \Leftrightarrow U = V$ by def'n
- Transitive: $U R V$ & $V R W \Rightarrow U R W$ by transitivity of subsets.

Example. For integers a, and b, define the relation D such that

$$a\,D\,b \Leftrightarrow a \mid b$$

Prove that D is a partial order relation.

- Reflexive: $a D a \Leftrightarrow a \mid a \Leftrightarrow a = ka$, $k \in \mathbb{Z}$
- Antisymmetric: Suppose $a D b$ and $b D a$. Then $b = k_1 a$ and $a = k_2 b$. Thus $b = k_1 a = k_1 k_2 b$, so $k_1 k_2 = 1$ and $k_1 = k_2 = 1 \Rightarrow a = b$
- Transitive: $a D b$ & $b D c \Rightarrow a D c$ by transitivity of divides

Definition.
The symbol $\preceq$ is often used to denote a general partial order relation.

Lexicographic Order
Let A be a set with a partial order relation R, and let S be a set of strings over A. Define a relation $\preceq$ on S as follows:

> Let s and t be any strings in S of length m and n, respectively, where m and n are positive integers, and let s_m and t_m be the characters in the mth position for s and t, respectively.

1. If $m \leq n$ and the first m characters of s and t are the same, then $s \preceq t$.
2. If the first $k-1$ characters in s and t are the same, $s_k \, R \, t_k$, and $s_k \neq t_k$, then $s \preceq t$.
3. If λ is the null string, then $\lambda \preceq s$.

If no strings are related by $\preceq$ other than by these three conditions, then $\preceq$ is a partial order relation on S.
This partial order relation is called the **lexicographic order for** S that corresponds to the partial order R on A.

Example. Let $A = \{x, y\}$ and let R be the following partial order relation on A:

$$R = \{(x, x), (x, y), (y, y)\}$$

Is $x \preceq x$?

Yes, by (1)

Is $x \preceq xx$?

Yes, by (1)

Is $yx \preceq yxy$?

Yes, by (1)

Is $xxxyyy \preceq xy$?

Yes, by (2)

Is $x \preceq y$?

Yes, by (2)

Is $\lambda \preceq xy$?

Yes, by (3)

Is $xyy \preceq xyx$?

No $\quad y \not R x$

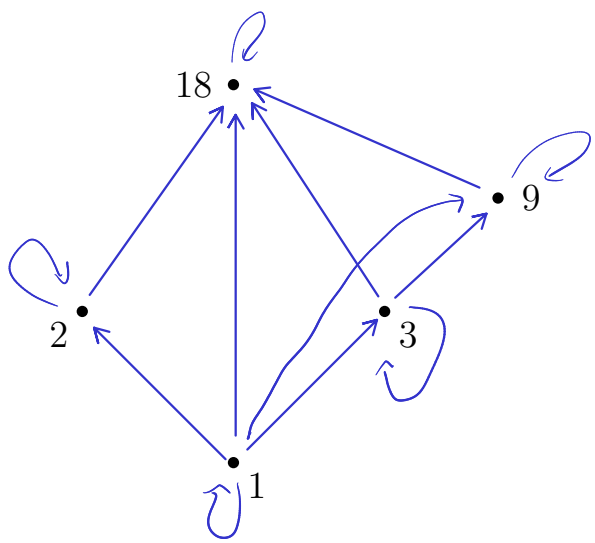
Definition.
A **Hasse diagram** is a graphical representation for a partial relation. To obtain a Hasse diagram, start with a directed graph of the relation with vertices placed so that all arrows point upwards. Then eliminate

- the loops at all vertices,
- all arrows implied by the transitive property, and
- the direction indicators on the arrows.

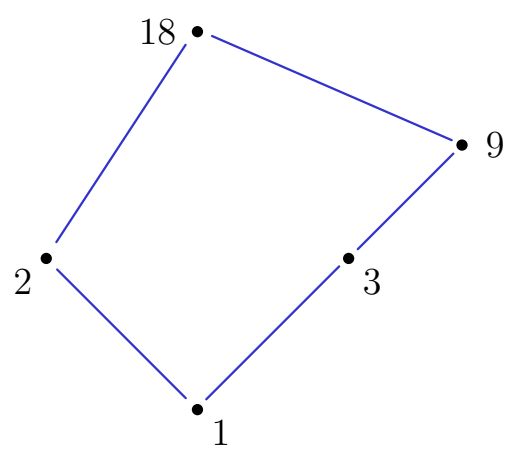
Example. Let $A = \{1, 2, 3, 9, 18\}$. Consider the partial order relation "divides":

$$a \, D \, b \Leftrightarrow a \mid b$$

Draw the directed graph of D:



Generate the Hasse diagram of D:

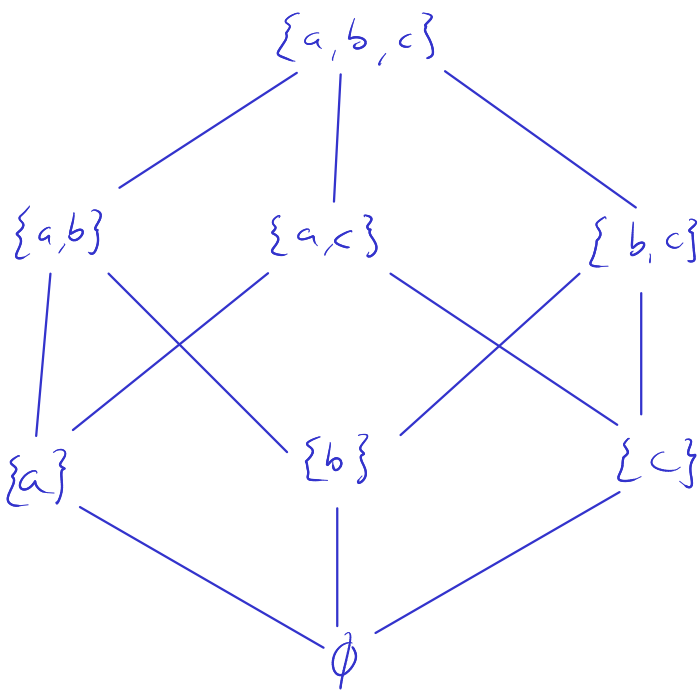


Example. Consider the "subset" relation, $\subseteq$, on the set $\mathscr{P}(\{a, b, c\})$:

$$U \, R \, V \Leftrightarrow U \subseteq V$$

Construct the Hasse diagram for this relation.

$\mathscr{P}(\{a,b,c\}) = \{\phi, \{a\}, \{b\}, \{c\}, \{a,b\}, \{a,c\}, \{b,c\}, \{a,b,c\}\}$



Definition.
Suppose $\preceq$ is a partial order relation on a set A. Elements a and b of A are said to be **comparable** if, and only if, either $a \preceq b$ or $b \preceq a$. Otherwise, a and b are called **noncomparable**.

When all the elements of a partial order relation are comparable, the relation is called a **total order**.

If R is a partial order relation on a set A, and for any two elements a and b in A either $a\,R\,b$ or $b\,R\,a$, then R is a **total order relation** on A.

A set A is called a **partially ordered set** (or **poset**) with respect to a relation $\preceq$ if, and only if, $\preceq$ is a partial order relation on A.

A set A is called a **totally ordered set** with respect to a relation $\preceq$ if, and only if A is partially ordered with respect to $\preceq$ and $\preceq$ is a total order.

A subset B of A is called a **chain** if, and only if, the elements in each pair of elements in B are comparable. The **length of a chain** is one less than the number of elements in the chain.

Note: The Hasse diagram for a total order relation can be drawn as a single vertical chain.

Example. The set $\mathscr{P}(\{a, b, c\})$ is partially ordered with respect to the subset relation. Find a chain of length 3 in $\mathscr{P}(\{a, b, c\})$.

$S = \{\phi, \{a\}, \{a, b\}, \{a, b, c\}\}$

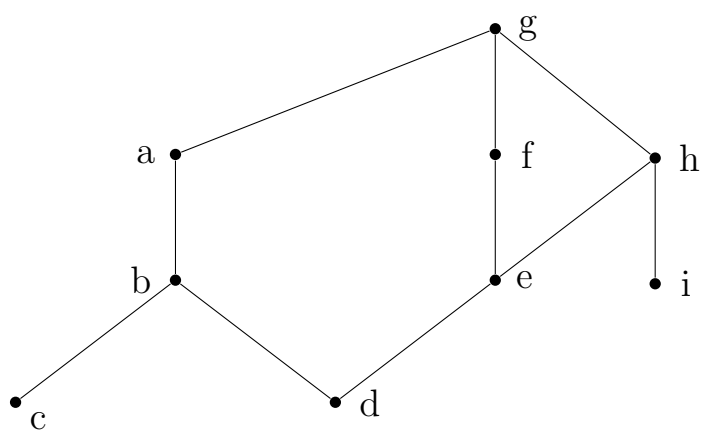
-or-

$S = \{\phi, \{c\}, \{a, c\}, \{a, b, c\}\}$

Definition.
Let a set A be partially ordered with respect to a relation $\preceq$. An element in A is called a

1. **maximal element of** $A \Leftrightarrow \forall b \in A$, either $b \preceq a$ or b and a are not comparable.
2. **greatest element of** $A \Leftrightarrow \forall b \in A$, $b \preceq a$.
3. **minimal element of** $A \Leftrightarrow \forall b \in A$, either $a \preceq b$ or b and a are not comparable.
4. **greatest element of** $A \Leftrightarrow \forall b \in A$, $a \preceq b$.

Example. Let $A = \{a, b, c, d, e, f, g, h, i\}$ have the partial ordering $\preceq$ defined by the following Hasse diagram. Find all maximal, minimal, greatest, and least elements of A.

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 Maximal:  g
Greatest:  g
 Minimal:  c,d,i
   Least:  N/A
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